Physical Distributed GNSS Spoofing Evades Relative-Distance Detection in Compact UAV Swarms

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[TODO: affiliazione, email]

***Abstract*—Multi-receiver spoofing detection is regarded as one of the most robust countermeasures against civilian GNSS spoofing: receivers in a known formation compare the mutual distances implied by their GNSS fixes with the formation geometry, and a single-transmitter spoofer—which forces every victim onto the same position—is detected with near-certain probability. We evaluate this detector family against a *physical distributed* spoofer: 24 fixed ground transmitters deployed around the operating area of a 25-UAV swarm, each replaying the satellite closest to it in azimuth, with capture decided by received power through an elevation-dependent ground-to-UAV channel and with the replayed code phase propagating over the true three-dimensional path to every drone. No per-victim signal shaping and no idealized attack are simulated. We implement the collapse detector of Jansen et al. (ACSAC 2016) in its three variants and a two-sided formation-consistency test representative of GNSS-vs-UWB checks for UAV swarms, and we evaluate them over the 5–20 m inter-UAV spacing of outdoor GPS-based formation flight, with 20 Monte Carlo missions per configuration and bootstrap confidence intervals. The attack displaces the swarm by 4 km while the computed formation is *dilated* by about 23 % rather than collapsed, so all three collapse detectors stay at $P_{\mathrm{D}} \leq 0.011$ (95 % upper bound) with ROC areas below chance (down to 0.001) at every spacing. The consistency test retains only marginal power, from $P_{\mathrm{D}} = 0.03$ at 5 m to 0.26 at 20 m at $P_{\mathrm{FA}} = 10^{-2}$, because the residual non-rigidity of ground transmitters—caused by the mismatch between satellite and transmitter elevations—is masked by GNSS noise at these spacings. We derive the residual to first order, show that it depends critically on deploying the array around the operating area, and delimit the result with respect to direction-of-arrival and absolute-reference defenses.**

***Index Terms*—GNSS spoofing, UAV swarms, spoofing detection, multi-receiver, distributed attack, relative distances, formation flight.**

I. Introduction

Civilian GNSS signals carry no authentication: their structure and navigation message are public, and a transmitter that reproduces them with enough power can make a receiver compute any position and time of the attacker's choosing. Since the first portable civilian spoofer was demonstrated [1], software-defined radios and open-source signal generators have made the attack accessible for a few hundred dollars, and incidents affecting ships, aircraft and unmanned systems are now reported routinely [2]. Unmanned aerial vehicles (UAVs) are a natural target: they navigate on GNSS by default, and a swarm of UAVs flying in formation—for surveying, inspection, delivery, or defense—multiplies both the value of the target and the consequences of a successful hijack.

Among the many countermeasures proposed in the last fifteen years [2], [3], *multi-receiver* detection stands out for its simplicity and for the strength of its guarantee. Its principle, established by Tippenhauer et al. [4], is geometric: a spoofer with a single antenna delivers the same set of signals to every receiver in its coverage area, so all the victims compute the *same* position fix. A group of receivers that knows its own formation can therefore detect the attack by checking that the mutual distances implied by the GNSS fixes have not collapsed. Jansen et al. [5] turned this check into a calibrated statistical test, showing that four receivers a few metres apart achieve a detection probability of 0.99 at a false-alarm rate of 10^{-5}; Swaszek and Hartnett [6] derived its Neyman–Pearson optimal form for commercial receivers. For UAV swarms the same idea has been proposed with the formation measured on-line by inter-drone ultra-wideband (UWB) ranging [7] or verified as a localization feasibility problem [8]. The guarantee rests on one assumption: that the attacker has a single transmission point.

This paper asks what happens when that assumption is dropped in the most concrete way possible—not with an idealized attacker that imposes a common offset on every victim, but with a *physical distributed* spoofer made of several fixed ground transmitters surrounding the swarm's operating area, each impersonating one satellite, subject to real propagation geometry and to a realistic ground-to-UAV channel. Such an attacker is within reach of a well-resourced adversary (two dozen synchronized software-defined radios), and recent work has shown formation-preserving spoofing to be feasible in hardware when the transmitters are placed along the satellite directions at altitude [9]. Ground-based transmitters, however, cannot reproduce the satellites' elevation, and the question of how much of the single-transmitter detection guarantee survives against them has not been quantified.

We answer it by simulation, for the inter-UAV spacing of 5 to 20 m that characterizes outdoor GPS-based formation flight (Section VI-A). Our contributions are:

- a transmitter-by-transmitter simulation model of a distributed ground spoofer attacking a moving UAV swarm, with per-epoch routing of satellites to transmitters, true three-dimensional propagation of the replayed signals, and the elevation-dependent ground-to-UAV path-loss model of [10] (Section IV);
- a faithful implementation of the collapse detector

of [5] in its three variants, and of a two-sided formation-consistency test representative of GNSS-vs-ranging checks [7], [11] (Section V);
- a Monte Carlo evaluation with mission-level bootstrap confidence intervals showing that the physical attack displaces the swarm by kilometres while *dilating*, not collapsing, the computed formation: the collapse detectors are driven below chance at every spacing, and the consistency test retains only marginal power ($P_{\mathrm{D}} \leq 0.26$ at $P_{\mathrm{FA}} = 10^{-2}$) (Section VI);
- a first-order analysis of the residual non-rigidity of a ground array, which identifies the satellite-vs-transmitter elevation mismatch as its cause, shows its dependence on the array deployment, and delimits the result with respect to direction-of-arrival and absolute-reference defenses (Sections IV-C and VII).

The simulator and the detection code are released with the paper.

II. Related Work

A. Multi-receiver detection in the position domain

Tippenhauer et al. [4] formalized the GPS group-spoofing problem and proved two complementary results: a single-antenna spoofer places all victims at the same location (Result 1), and a formation-preserving placement of one transmitter per impersonated satellite always exists, along the satellite directions (Result 7). Their countermeasure checks that the computed positions preserve the physical formation, without a statistical model. Jansen et al. [5] supplied the model: the pairwise distances between receivers are Rician under authentic conditions and Rayleigh under a single-transmitter attack, with spatially correlated position noise ($\sigma \approx 0.5$–$3\,\mathrm{m}$, $\rho \approx 0.4$–0.6 measured), and the test $f(\{d_{ij}\}) < \lambda$ is evaluated in minimal, maximal, majority and normalized-majority variants, the last being the best. Their prototype with four receivers on a 5 m square ran for three hours without errors; their own attempt at a distance-preserving four-USRP attack succeeded on one receiver and failed on two. Swaszek and Hartnett [6] derived Neyman–Pearson optimal tests for COTS receivers at known relative positions, and later fused GNSS fixes with inter-vehicle ranges for platoons under the assumption that only one vehicle is spoofed [12]. Crowd-GPS-Sec [13] applies the collapse principle to crowdsourced air-traffic receivers, combined with an absolute time-difference-of-arrival reference.

B. Measurement-domain and swarm variants

In the measurement domain, Stenberg et al. [11] test double differences of pseudorange and carrier phase between receivers with a χ^2 GLRT, finding a minimum receiver separation of about 5 m with standard positioning; Kalantari and Larsson [14] derive a GLRT for receivers on a rigid body with unknown pose explicitly against multiple spoofers, showing that more receivers make the attack harder; Borio and Gioia [15] detect the single common direction of arrival with a two-receiver carrier-phase sum-of-squares statistic. For UAV swarms, Mykytyn et al. [7] compare GNSS-derived inter-drone distances with UWB two-way ranging and raise an alarm when they differ by more than about 5 m, dismissing a coordinated per-drone attack as "virtually impossible"; Bi et al. [8] cast detection as a semidefinite localization-feasibility problem robust to colluding members. GNSS/INS fusion defenses such as SemperFi [16] and tightly coupled INS innovation monitors [17], [18] compare the GNSS track with inertial dead reckoning and respond to the time variation of the induced error.

C. Coordinated and distributed spoofing

Attack-side work has progressively approached the formation-preserving placement of [4]. Ceccato et al. [19] designed a multi-antenna ground spoofer with Wiener precoding that displaces a swarm without disrupting its formation, at the level of a mathematical model; Chen et al. [20] analyze formation-preserving spoofing of small swarms with software-defined radios; GNSS-WASP [9] demonstrated in hardware a constellation of at least four GPSDO-synchronized transmitters, placed in line with the satellites at altitude, that spoofs an arbitrary number of receivers over a square kilometre while preserving their relative distances to within a metre, explicitly bypassing multi-receiver [5] and sensor-fusion [16] countermeasures. Park and Yoo [21] state the translation-invariance of relative-only detectors as a formal proposition, rooted in the gauge freedom of anchor-free network localization [22], and evaluate it by simulation with an idealized common offset. Surveys of UAV spoofing defenses [3], [23] note that detectors relying on neighbours' positions fail when the neighbours are inside the spoofed area, but none quantifies the effect for a physical ground array.

D. Positioning of this work

Our attacker is deliberately more constrained than those of [9], [21]: fixed transmitters on the ground, no per-victim adaptation, a measured channel. The quantity we report is precisely what separates it from the ideal formation-preserving attack—the residual deformation of the computed formation—and its consequence for the two families of relative-distance detectors over the spacing range of real formation flight.

III. System Model

A. Swarm, constellation, and measurement model

We consider a swarm of $m = 25$ UAVs flying in a square 5×5 grid with spacing g (the *formation spacing*), following a way-point trajectory at altitudes between 100 and 700 m that spans about 7 km around its start point; positions are sampled every $T_s = 5\,\mathrm{s}$ over a 20-minute mission. The GNSS constellation is modeled as $N_{\mathrm{sat}} = 24$ satellites on 6 circular orbital planes (inclination $55°$, altitude 20 200 km) over a rotating spherical Earth. For UAV u at position $\mathbf{p}_u(t)$ and satellite s at position $\mathbf{s}_s(t)$ (visible when its elevation is positive), the authentic propagation delay is $\tau_{u,s}(t) = \|\mathbf{s}_s(t) - \mathbf{p}_u(t)\|/c$. Each receiver has a clock bias $\delta t_u \sim \mathcal{U}(-1, 1)\,\mathrm{s}$, constant

over the mission, and each pseudorange carries additive white Gaussian noise of standard deviation $\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$, representative of standard (non-differential) positioning.

Each UAV computes its position fix $\hat{\mathbf{p}}_u$ independently by iterated linearized least squares (Gauss–Newton) over the four unknowns $(x, y, z, \delta t)$: at iteration k, with predicted ranges $R_s = \|\mathbf{s}_s - \hat{\mathbf{x}}^{(k)}\|$ and pseudoranges $\rho_s = c\,(\tilde{\tau}_{u,s} - \widehat{\delta t}^{\,(k)})$,

$$\mathbf{G}\,\Delta = \boldsymbol{\rho} - \mathbf{R}, \qquad \mathbf{G}_{s,:} = \left[\frac{(\hat{\mathbf{x}} - \mathbf{s}_s)^{\top}}{R_s},\ c\right], \tag{1}$$

solved in the least-squares sense for five iterations, where $\tilde{\tau}_{u,s}$ is the *measured* delay (authentic or spoofed). Without attack and noise the solver returns the true grid positions to numerical precision (error $< 10^{-4}\,\mathrm{m}$), so estimated and true positions are expressed in the same frame.

In addition to GNSS, the UAVs measure their instantaneous mutual distances with an independent on-board ranging sensor (UWB two-way ranging, as in [7]):

$$d_{ij}^{\mathrm{meas}}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| + \varepsilon_{ij}(t), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{\mathrm{rng}}^2), \tag{2}$$

with $\sigma_{\mathrm{rng}} = 0.1\,\mathrm{m}$. This is the most favorable setting for the defender: the true formation is known essentially without error at every epoch.

B. Ground-to-UAV channel

Spoofing signals travel over a ground-to-air link. We adopt the suburban cellular-to-UAV path-loss model of Al-Hourani and Gomez [10]: for a ground transmitter at horizontal distance d and elevation angle θ (in degrees) as seen from the UAV,

$$\mathrm{PL}(d, \theta) = 10\,\alpha \log_{10}(d) + A\,(\theta - \theta_0)\,e^{-\frac{\theta - \theta_0}{B}} + \eta_0 \quad [\mathrm{dB}], \tag{3}$$

with $\alpha = 3.04$, $A = -23.29$, $B = 4.14$, $\theta_0 = -3.61^{\circ}$, $\eta_0 = 20.70\,\mathrm{dB}$ (shadowing disabled for repeatability). The excess-loss term is fitted in [10] for $\theta \in [-2^{\circ}, 10^{\circ}]$ and is saturated at 10° beyond; with the deployment of Section IV no link exceeded this limit in any run, so the model is always used within its validity range. Authentic signals follow free-space loss from satellite transmit power $P_t^{\mathrm{sat}} = 25\,\mathrm{W}$ with antenna gains $G_t^{\mathrm{sat}} = 13.5\,\mathrm{dB}$ and $G_r = 5\,\mathrm{dB}$, giving received powers of about $-150\,\mathrm{dBW}$.

IV. Physical Distributed Spoofing Attack

The attacker's goal is to displace the whole swarm's perceived position onto a decoy point about 5.7 km from the flight path, while the swarm believes it is following its mission plan. The attack is simulated transmitter by transmitter; nothing is assumed about per-victim signal shaping.

A. Transmitter geometry and per-epoch routing

$N_{\mathrm{sp}} = 24$ fixed ground transmitters are placed on a circle of radius $R_{\mathrm{sp}} = 10\,\mathrm{km}$ enclosing the swarm's operating area (Fig. 1), each radiating $P_t^{\mathrm{sp}} = 1\,\mathrm{mW}$ with $G_t^{\mathrm{sp}} = 1\,\mathrm{dB}$. Following the geometric argument of [4, Result 7], each visible satellite s is routed, at every epoch, to the transmitter whose azimuth as seen from the *current* true swarm centre $\mathbf{p}^{\mathrm{ctr}}(t)$ is closest to the satellite's azimuth:

$$\mathrm{sp}(s, t) = \arg\min_{k} \left|\angle_{2\pi}\big(\phi_s(t) - \phi_k(t)\big)\right|. \tag{4}$$

The transmitters never move; only the routing is updated, which requires the attacker to track the swarm centre—the same knowledge it needs to calibrate the replayed delays.

B. Capture and replayed delay

For each UAV u and visible satellite s the receiver locks onto the transmitter whenever the received spoofing power, computed with (3), exceeds the authentic one. With the chosen EIRP the spoofer margin over the satellite is between 13 and 29 dB on every link (median 19.5 dB), so capture is complete and homogeneous across the swarm: no UAV mixes authentic and spoofed satellites.

Transmitter $k = \mathrm{sp}(s, t)$ replays the signal of satellite s with its code phase calibrated so that, at the true swarm centre, the measured delay equals the decoy delay τ_s^{dec}—the delay that satellite s would exhibit at the decoy position. A UAV at a different position receives the same transmission over a different path and measures

$$\tilde{\tau}_{u,s} \;=\; \tau_s^{\mathrm{dec}} \;-\; \delta t_u \;+\; \frac{\|\mathbf{q}_k - \mathbf{p}_u\| - \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|}{c}, \tag{5}$$

where $\mathbf{q}_k$ is the transmitter position and both norms are true three-dimensional slant distances. The last term is the physical propagation residual: it vanishes for the centre UAV, which computes exactly the decoy position, and is nonzero for the other UAVs.

C. Residual non-rigidity of a ground array

Write $\mathbf{p}_u = \mathbf{p}^{\mathrm{ctr}} + \boldsymbol{\delta}_u$ with $\|\boldsymbol{\delta}_u\| \ll \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|$. To first order the residual in (5) is $-(\hat{\mathbf{w}}_k \cdot \boldsymbol{\delta}_u)/c$, with $\hat{\mathbf{w}}_k$ the unit vector from the centre to the transmitter, whereas the delay that would make UAV u compute the decoy position *rigidly displaced* by $\boldsymbol{\delta}_u$ is $-(\hat{\mathbf{e}}_s \cdot \boldsymbol{\delta}_u)/c$, with $\hat{\mathbf{e}}_s$ the unit line of sight to the satellite. For a horizontal swarm and azimuth-matched routing the per-satellite pseudorange error is

$$\epsilon_{u,s} \;\approx\; (\hat{\mathbf{a}}_s \cdot \boldsymbol{\delta}_u)\,\big(\cos\theta_k - \cos\mathrm{el}_s\big), \tag{6}$$

where $\hat{\mathbf{a}}_s$ is the satellite azimuth direction, θ_k the near-zero elevation of the ground transmitter and el_s the satellite elevation. A ground transmitter cannot reproduce the satellite's elevation, so (6) vanishes only for satellites near the horizon. Its coefficient is positive for every satellite: a UAV displaced *towards* satellite s measures a pseudorange longer than the rigid one, so the least-squares solver places it farther from the centre than it is. The error scales with the baseline and the computed formation is therefore *dilated*—a systematic outward displacement of every UAV, modulated by the elevations of the satellites in view—rather than collapsed. Section VI measures the dilation at about 23 % of the baseline.

Two further facts, established numerically, matter for an attacker. The residual does not depend on transmitter power: capture is already complete at 1 mW and raising the power to

10 W leaves the deformation unchanged. It depends critically on the array enclosing the operating area: with a 2.5 km array centred on the start point the swarm is inside the circle only 26 % of the time, and the deformation grows from 12 % inside to 55 % outside (correlation 0.98 with the distance from the array centre), because from outside the array all transmitters lie in a narrow angular sector and no routing can match the satellite azimuths.

Remark 1 (Why collapse detectors are blind): The collapse detectors of Section V threshold a statistic that decreases when the pairwise distances shrink—the signature of a single transmitter seen by all receivers [4, Result 1]. A distributed spoofer with one transmitter per satellite delivers to every UAV a different set of delays, consistent to first order with its own displacement from the centre, and the residual (6) dilates the formation. The statistic therefore moves *away* from its alarm region, and no threshold can recover the attack.

V. Relative-Distance Detection Methods

All detectors operate on the normalized distance ratios

$$r_{ij}(t) = \frac{\hat{d}_{ij}(t)}{d_{ij}^{\mathrm{meas}}(t)}, \qquad \hat{d}_{ij} = \|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\|, \tag{7}$$

computed over all $\binom{m}{2} = 300$ pairs at every epoch, where d_{ij}^{meas} is the instantaneous UWB measurement of (2); under authentic conditions $r_{ij} \approx 1$.

A. Collapse tests (Jansen et al.)

The ACSAC 2016 detector [5] declares spoofing when the pairwise distances *shrink*:

$$f\big(\{r_{ij}\}\big) \underset{H_1}{\overset{H_0}{\gtrless}} \lambda, \qquad \text{alarm if } f < \lambda, \tag{8}$$

with the three variants of its Appendix A implemented here:

$$f_{\min} = \min_{i<j} r_{ij}, \quad f_{\max} = \max_{i<j} r_{ij}, \quad f_{\mathrm{maj}} = \operatorname*{med}_{i<j} r_{ij}, \tag{9}$$

where the median form is equivalent to the paper's normalized majority voting: at least half of the pairs below λ iff the median is. Jansen et al. set λ from a Rician (H_0) / Rayleigh (H_1) model of the pairwise distance with spatially correlated position noise, $\sigma_d = \sqrt{2}\,\sigma\sqrt{1-\rho}$; we calibrate λ empirically to the same effect (Section V-C).

B. Formation consistency test

As a representative of the two-sided GNSS-vs-ranging consistency family — $|d_{\mathrm{GPS}} - d_{\mathrm{UWB}}| > d_{\mathrm{thr}}$ for UAV swarms [7], double-difference GLRTs across receivers [11], GLRTs fusing GNSS fixes with inter-vehicle ranges [12] — we use the worst-pair statistic

$$T_{\mathrm{cons}} = \max_{i<j} \big|\, r_{ij} - 1 \,\big|, \qquad \text{alarm if } T_{\mathrm{cons}} > \lambda. \tag{10}$$

Unlike (8), this test fires on *any* deformation of the formation, in either direction, and is therefore the natural candidate against a distributed attacker.

C. Threshold calibration and metrics

Following the Neyman–Pearson practice of [5], thresholds are calibrated on the H_0 (no-spoofing) scenario simulated with the same noise levels: λ is the empirical P_{FA}-quantile (lower for the collapse tests, upper for the consistency test) of the statistic over all H_0 epochs, with $P_{\mathrm{FA}} = 10^{-2}$. Performance is reported as (i) the detection probability P_{D}, the fraction of attacked epochs that raise an alarm; (ii) ROC curves and their area (AUC) obtained by sweeping λ, oriented so that AUC $= 0.5$ is chance and AUC < 0.5 means the attack moves the statistic away from the alarm region; and (iii) the median relative distance distortion $E = \mathrm{med}_{i<j}\,|\hat{d}_{ij} - d_{ij}|/d_{ij}$, the geometric evidence available to any relative-distance detector. Because the epochs of one mission share clocks, trajectory and satellite geometry, the Monte Carlo unit is the *mission*: 95 % confidence intervals on P_{D} are obtained by a paired bootstrap over missions (500 resamples; H_0 and attack of a mission share the random seed, and the threshold is re-calibrated in every resample).

VI. Simulation Results

A. Setup and formation-spacing regime

Table I summarizes the parameters. We sweep the formation spacing over $g \in \{5, 8, 10, 12, 15, 20\}$ m. This range is the one of outdoor formation flight on GPS: experimental leader–follower formations of quadcopters are flown at 5 m spacing, with separation errors of 0.4 m [24] to 1.4–1.9 m RMSE even with RTK corrections [25]; a design method that derives the minimum safe spacing from the combined navigation and control error yields spacings of metres for GNSS-based navigation [26]; the multi-receiver prototype of [5] uses a 5 m square, and the swarm detector of [7] is designed around a 5 m GPS error budget. Closer spacings—down to 0.8 m in drone light shows [27]—require centimetre-level RTK positioning and are outside the standard-GPS regime considered here, while wider spacings belong to coverage and surveillance swarms rather than to formation flight. For each spacing we simulate $N_{\mathrm{MC}} = 20$ independent missions (random start time, clock biases and noise), each of 240 epochs, with and without the attack. The simulator and detection code are available in the companion repository.[1]

B. Attack effectiveness

The attack succeeds: the centre UAV's estimated position is dragged by the full decoy offset, 4.0 km on average over the mission, at every epoch (Fig. 2, top), and every UAV in the swarm is captured on all its visible satellites. The bottom panel shows the price paid by a ground array: the relative-distance distortion is raised above the no-attack noise floor throughout the mission.

[1] https://github.com/silviasch01/GPS_spoofing_distribuito

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
Swarm size m	25 (5×5 grid)
Formation spacing g	5, 8, 10, 12, 15, 20 m
Mission duration / epoch	1200 s / $T_s = 5$ s
Satellites	24 (6 planes, 55°, 20 200 km)
Pseudorange noise σ_{pr}	3 m
Ranging noise σ_{rng}	0.1 m
Ground transmitters N_{sp}	24 on circle, $R_{\mathrm{sp}} = 10$ km
Transmitter EIRP	1 mW + 1 dB
Ground-to-UAV channel	Eq. (3), [10]
Spoofer power margin (all links)	13.2–29.0 dB
Target false-alarm rate P_{FA}	10^{-2}
Monte Carlo missions per spacing	20

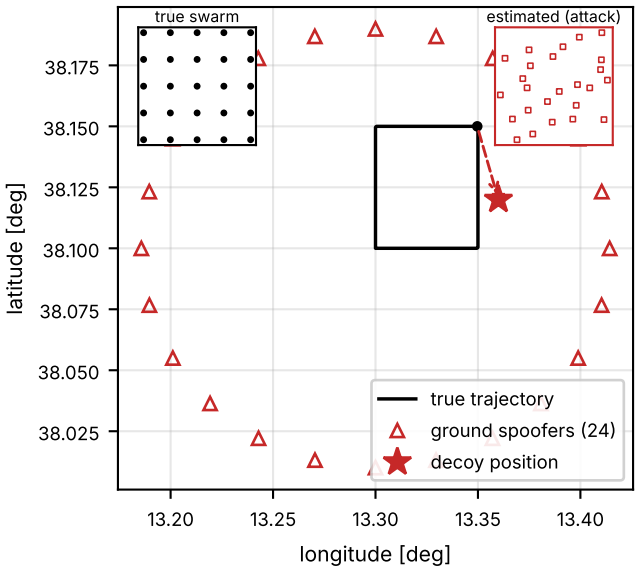


Fig. 1. Attack scenario (one epoch, $g = 10$ m). The swarm (true positions, dots) flies its mission inside a 10 km circle of 24 fixed ground transmitters (triangles), each replaying one satellite. The estimated fixes (open squares) are displaced onto the decoy (star) and mildly dilated.

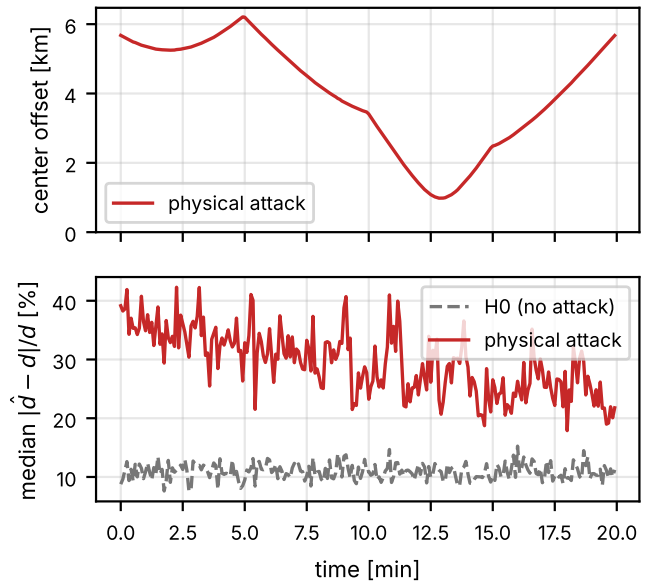


Fig. 2. Per-epoch behaviour at $g = 10$ m (one mission). Top: position offset of the centre UAV. Bottom: median relative-distance error with and without the attack.

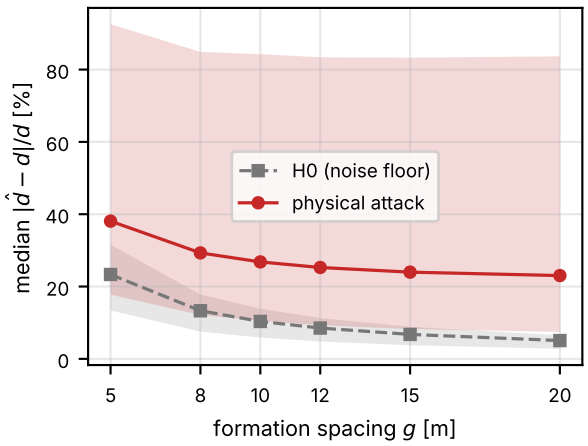


Fig. 3. Median relative-distance distortion vs. formation spacing (median over 20 missions; band: 10th–90th percentile across missions).

C. Formation distortion and dilation

Figure 3 reports the median relative distance distortion E over the 20 missions (band: 10th–90th percentile across missions). Under H_0 the distortion is pure GNSS noise and decays as $1/g$, from 22 % at $g = 5$ m to 4.6 % at 20 m. Under attack it decreases from 38 % at 5 m to 23 % at 20 m, i.e. towards the scale-free plateau of the residual (6), whose magnitude depends on the elevations of the satellites in view and therefore varies between missions. The sign of the residual is what Remark 1 predicts: the median distance ratio $f_{\mathrm{maj}} = \mathrm{med}\, r_{ij}$ is 1.23 under attack at $g = 20$ m against 1.01 without attack (and 1.37 against 1.16 at 5 m, where noise biases both upward), and it exceeds one in every attacked epoch. The computed formation is dilated by about a quarter of the baseline; it never collapses.

D. Detection performance

Table II and Fig. 4 report P_{D} at $P_{\mathrm{FA}} = 10^{-2}$ with 95 % confidence intervals, Table III the ROC areas, and Fig. 5 the ROC curves at $g = 10$ m. Three observations follow.

First, **the collapse detectors are defeated at every spacing**: all three Jansen variants give $P_{\mathrm{D}} \leq 0.011$ (95 % upper bound), at or below the false-alarm rate, and their AUC is *below* chance—down to 0.001 for the majority variant at 20 m, i.e. the attacked statistic is larger than the authentic one in essentially every epoch. This is the consequence of Remark 1: the distributed attack dilates the formation, so the one-sided statistic of (8) is pushed away from its threshold; the majority variant, the one recommended in [5], never fires at all. No

TABLE II

DETECTION PROBABILITY AT $P_{\mathrm{FA}} = 10^{-2}$ AGAINST THE PHYSICAL DISTRIBUTED ATTACK ($\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$, $\sigma_{\mathrm{rng}} = 0.1\,\mathrm{m}$; 20 MISSIONS, 4800 EPOCHS PER CELL; 95 % CI FROM THE PAIRED MISSION BOOTSTRAP).

g [m]	Jansen min	Jansen max	Jansen majority	Consistency
5	0.005 [0.004, 0.011]	0.007 [0.004, 0.009]	0.000 [0.000, 0.000]	0.029 [0.016, 0.048]
8	0.002 [0.000, 0.004]	0.003 [0.002, 0.005]	0.000 [0.000, 0.000]	0.065 [0.030, 0.117]
10	0.002 [0.000, 0.004]	0.002 [0.001, 0.005]	0.000 [0.000, 0.000]	0.113 [0.039, 0.202]
12	0.001 [0.000, 0.004]	0.001 [0.000, 0.003]	0.000 [0.000, 0.000]	0.153 [0.058, 0.264]
15	0.002 [0.000, 0.005]	0.001 [0.000, 0.001]	0.000 [0.000, 0.000]	0.193 [0.085, 0.310]
20	0.001 [0.000, 0.003]	0.000 [0.000, 0.001]	0.000 [0.000, 0.000]	0.262 [0.130, 0.423]

TABLE III

ROC AREA (AUC) AGAINST THE PHYSICAL DISTRIBUTED ATTACK; 0.5 IS CHANCE, VALUES BELOW 0.5 MEAN THE ATTACK MOVES THE STATISTIC AWAY FROM THE ALARM REGION.

g [m]	Jansen min	Jansen max	Jansen maj.	Consistency
5	0.312	0.430	0.171	0.570
8	0.262	0.378	0.072	0.622
10	0.221	0.349	0.037	0.651
12	0.189	0.324	0.019	0.676
15	0.156	0.293	0.006	0.706
20	0.122	0.255	0.001	0.743

threshold choice changes this.

Second, **the consistency test retains only marginal power**: P_{D} rises from 0.03 at 5 m to 0.26 at 20 m, with AUC from 0.57 to 0.74 and wide confidence intervals that reflect the mission-to-mission variability of the residual. The test sees the dilation only where it exceeds the $1/g$ noise floor of standard positioning; over the whole formation-flight regime, three attacked epochs out of four raise no alarm even at the widest spacing, and at 5–10 m the attack passes essentially undetected.

Third, **the residual is the only handle the defender has**. It is not a tuning artifact of the attack (power does not change it), it is not removable by routing once the array encloses the operating area, and it is the geometric cost of keeping the transmitters on the ground; its magnitude is governed by the satellite elevations through (6).

VII. DISCUSSION AND LIMITATIONS

Scope of the result: The evaluation covers detectors that are functions of relative position/distance quantities—the collapse family and the two-sided consistency family—over the spacing of GPS-based formation flight. The former is blind for a structural reason (Remark 1) that does not depend on the spacing; the latter is limited by the ratio between the ground-array residual and the GNSS noise, and would detect the attack for wide formations, for low-noise (RTK) fixes, or with longer integration windows—our earlier runs at 50 and 100 m gave $P_{\mathrm{D}} = 0.56$ and 0.65. The result does *not* extend to (i) *direction-of-arrival* detectors: a circle of ground transmitters produces arrivals at near-zero elevation, incongruent with the satellite constellation, so multi-antenna arrays with attitude information [2], [15] would see our attacker, unless its transmitters are placed along the satellite directions as in [9]; nor to

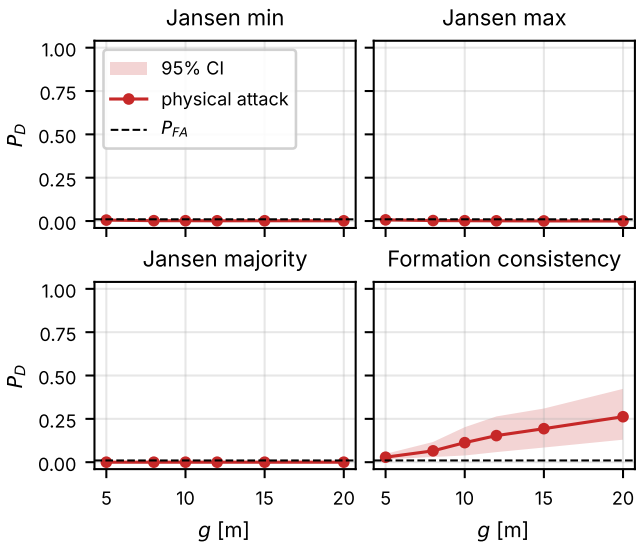


Fig. 4. Detection probability vs. formation spacing at $P_{\mathrm{FA}} = 10^{-2}$ for the four detectors (20 missions; band: 95 % bootstrap CI). The collapse detectors never leave the P_{FA} line.

(ii) *absolute-reference* checks—ranging to fixed infrastructure, TDoA from ground sensor networks [13], terrain or signal-of-opportunity positioning—for which the 4 km displacement is directly observable; nor to (iii) GNSS/INS detectors such as SemperFi [16] and INS innovation monitors [17], [18], which respond to the time variation of the induced error and were not evaluated.

Relation to formation-preserving attacks: Tippenhauer et al. [4] prove that a formation-preserving placement exists and GNSS-WASP [9] realizes it with transmitters in line with the satellites at altitude; Park and Yoo [21] show that an exactly common translation is invisible to any relative-only detector. Our attacker is deliberately more constrained, and the quantity we measure is what separates it from those ideal attacks: a dilation of about 23 % of the baseline, enough to be seen by a two-sided test on wide formations and too small to be seen on the compact ones of formation flight. Raising the transmitters would reduce the residual according to (6); we leave that trade-off to future work.

Deployment and steady state: The result assumes an array that encloses the swarm's operating area (Section IV-C); a

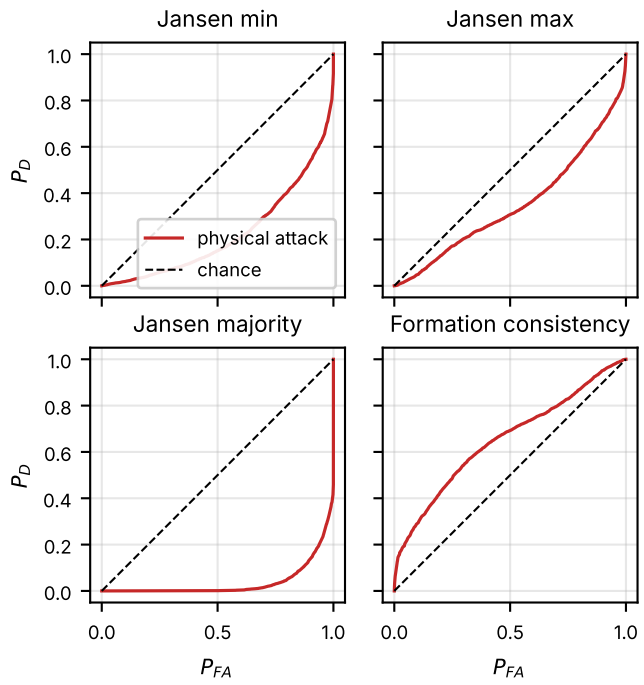


Fig. 5. ROC curves at $g = 10\,\mathrm{m}$ against the physical distributed attack. The collapse detectors operate below chance.

static array that the swarm leaves deforms the formation by more than 50 % and is readily detected. The decoy position is constant, so the attack is modeled in its steady state after capture; the transient at capture and time-varying decoys are not modeled.

Other simplifications: The simulator uses a spherical Earth, a synthetic constellation, uncorrelated pseudorange noise (spatial correlation would lower the detector's noise floor [5] and favour the consistency test), no multipath or shadowing, and signal capture by power comparison without tracking-loop dynamics. Twenty missions per spacing bound P_{D} to within about ± 0.1 for the consistency test; the conclusions on the collapse detectors do not depend on this resolution.

VIII. Conclusion

We evaluated the classical multi-receiver relative-distance spoofing detectors against a physical distributed ground spoofer attacking a UAV swarm in the 5–20 m spacing of outdoor GPS-based formation flight, simulating the attack transmitter by transmitter without any idealization. The attack displaces the swarm by 4 km and defeats the collapse detectors of Jansen et al. in all their variants at every spacing ($P_{\mathrm{D}} \leq 0.011$, AUC below chance), because a distributed spoofer dilates rather than collapses the computed formation and so never produces the signature these detectors rely on. A two-sided consistency check against independent UWB ranging retains only marginal power (P_{D} from 0.03 to 0.26), limited by a residual dilation of about 23 % of the baseline that is the geometric cost of ground-based transmitters and is masked by standard-GPS noise at these spacings. Formation flight on GNSS therefore cannot rely on relative-distance checks alone: they should be complemented by direction-of-arrival or absolute-reference information, which the same residual analysis identifies as the quantities a ground array cannot forge.

Acknowledgments

[TODO: ringraziamenti, funding]

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